A-A-53823
20 February 1989

COMMERCIAL ITEM DESCRIPTION

BUR, DENTAL, FINISHING (Steel)

The General Services Administration has authorized the use of this Commercial Item Description in preference to Federal Specification GG-B-001275 dated 17 October 1967.

Salient characteristics.

This commercial item description covers angle latch finishing burs suitable for use in finishing dental restorations.

The finishing burs shall be of the following shapes and size numbers as specified below.

Interproximal	No. 6
Round	No. 200
Oval	No. 218
Sugar-loaf	No. 242
Round	No. 012
Round	No. 016

Material. The burs shall be fabricated from steel having a minimum tungsten content of 3.00 percent, and a minimum carbon content of 1.00 percent. As an alternate, the burs shall be fabricated from steel having a minimum tungsten content of 0.90 percent, a minimum vanadium content of 0.08 percent and a minimum carbon content of 1.00 percent.

Style, design and dimensions. The burs shall consist of a shank, neck, and head section.

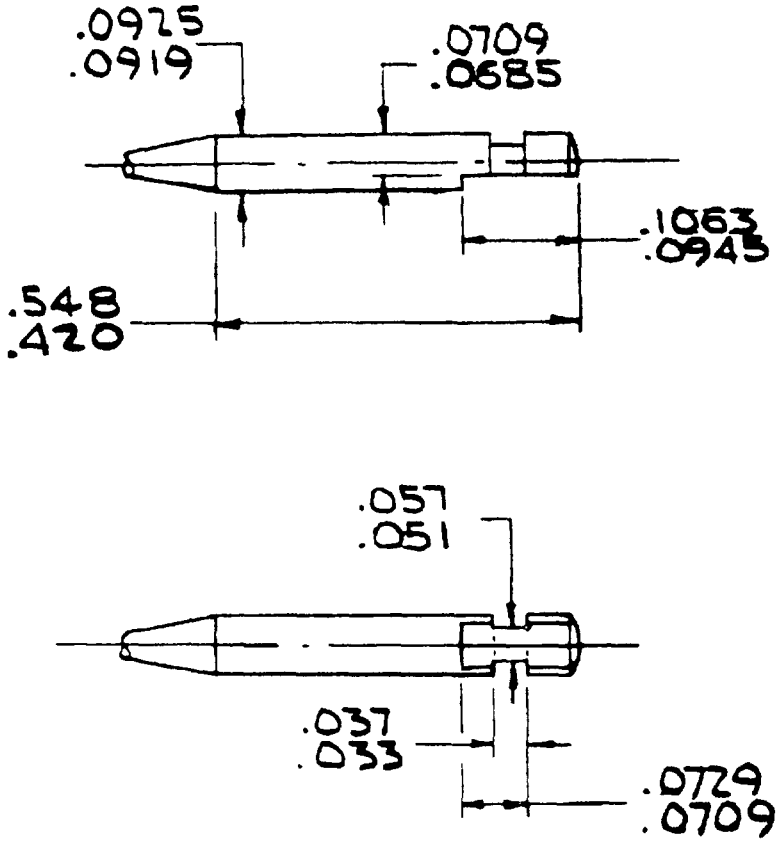
Shank. The shank dimensions shall be as specified in figure 1. The overall length of the burs shall be 0.875 ± .010 inch.

Head and neck. The head and neck of the burs shall be straight and symmetrical. The blades shall be uniformly cut and sharp. The head and neck dimensions and the number of blades for the burs, shall be as specified in table I for the applicable shape and size number. The shape of the bur heads listed in table I shall be as shown in figures 2 through 5.

AMSC N/A FSC 6520

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NOTE:
ALL DIMENSIONS ARE IN INCHES

FIGURE 1. DIMENSIONS OF SHANK END OF BUR

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FIG 2



FIG 3

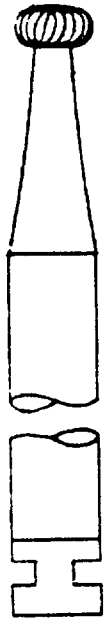


FIG 4



FIG 5

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TABLE I. Bur head and neck data.

Fig. No.	Shape	Size Number	No. of Blades (± 1)	Length of head inch /1	Diameter of head, inch /2	Diameter of neck, inch maximum
2	Inter-proximal	6	18	0.216	0.0715	0.055 to 0.059 /3
3	Round	200	20	-----	0.081	0.040 to 0.044
	Round	012	15	-----	0.047	0.035 max.
	Round	016	17	-----	0.062	0.041 max.
4	Oval	218	20	0.038	0.081	0.040 to 0.044
5	Sugarloaf	242	12	0.170	0.039	0.029 to 0.032

/1 The tolerance on the length of head of the interproximal bur shall be ± 0.008 inch. Head lengths for the other burs are minimums.

/2 The head diameter shall be designated at the maximum cross-section on the head. The tolerance for the interproximal bur shall be ± 0.0035 inch. The tolerance for all the other burs shall be ± 0.003 inch.

/3 The interproximal bur shall have a neck length of 0.189 to 0.205 inch.

Hardness. The hardness of the bur heads shall be no less than 750 on the Vickers scale (Diamond pyramid hardness) when tested as specified herein. The bur necks shall be properly tempered to resist fracture or bending in use.

Runout. The total indicated runout of the bur shall not exceed 0.003 inch, when tested as specified herein.

Finish. The bur head shall have a finely machined finish and the blades properly sharpened. The shank and neck shall have a smooth finish.

Identification. Each bur shank may be stamped with the name or registered trademark of the manufacturer. Such marking shall be permanent and legible. Identification markings on burs shall be at the option of the manufacturer or contractor.

Workmanship. The burs shall be free from defects which detract from their appearance or may impair their serviceability, proper function and intended use.

Unit. Package (PG). One package containing six burs, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality for raw material, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 at inspection level II, except for dimensions which shall be inspection level S-1, AQL of 2.5 (percent defective). The unit of product for examination purposes shall be one bur.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 at inspection level S-1, AQL 2.5 (percent defective), except for material and hardness the AQL shall be zero for all sample sizes.

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Examination. The burs shall be examined for defects as listed below, but the examination shall not be restricted to those defects below.

Defects

Burs not of shape and size specified.
Shank design and dimensions not as specified.
Shapes and dimensions of head and neck not as specified.
Blades not properly cut; not sharp and free from nicks.
Finish of bur not as specified.
Latch end of burs not clean machined.
Overall lengths not as specified.
Packaging, packing, and marking not as specified.
Marking on burs missing or improper.

Dimensions. Examination of the dimensions, in figure 1, shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturers' gages may be used if properly calibrated checking standards are used in conjunction with them.

Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a miscroscope of 30 power (30X) magnification.

Tests.

Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the burs shall be determined using method No. 111 or 112 of FED-STD-151.

Hardness. The Vicker hardness test shall be in accordance with ASTM E92.

Runout. The total indicated runout of the bur neck shall be measured to the nearest 0.0005 inch. The bur shall be held in a vee-block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the bur and shall be so positioned that the foot contacts the bur neck at the point of smallest diameter. The difference between the maximum and minimum readings for one full rotation of the bur shall be taken as the total indicated runout.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquistion Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Numbers are covered by this document:

NSN	Item Identification	Specification Designation Size No.	Shape
6520-00-299-9677	BUR, DENTAL, FINISHING, Angle Handpiece, Steel No. 6s 69	6	Inter-proximal
6520-00-510-0650	BUR, DENTAL, FINISHING, Angle Handpiece, No. 200, 6s	200	Round
6520-00-510-1050	BUR, DENTAL, FINISHING, Angle Handpiece, Steel, No. 218, 6s	218	Oval
6520-00-510-2550	BUR, DENTAL, FINISHING, Angle Handpiece, Steel, No. 242, 6s	242	Sugar-loaf
6520-01-105-4107	BUR, DENTAL, FINISHING, Angle Handpiece, Steel, No. 012, 6s	012	Round
6520-01-105-4108	BUR, DENTAL, FINISHING Angle Handpiece, Steel, No. 016, 6s	016	Round

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2207

LOCATION: NCR Disk No. 7

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